Exercise 3.02

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

part a

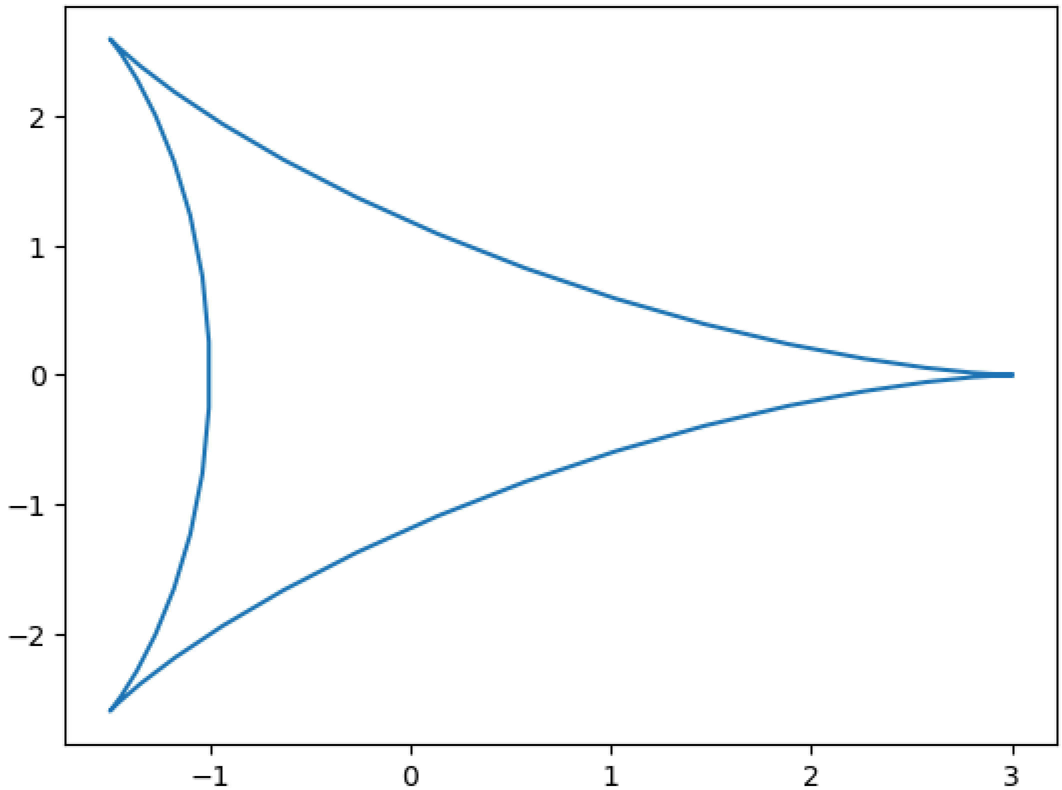
In [2]:
```
theta = np.linspace(0,2*np.pi)

x = 2*np.cos(theta)+np.cos(2*theta)
y = 2*np.sin(theta) - np.sin(2*theta)
```

In [5]:
```
y
```

Out[5]:
```
array([ 0.00000000e+00,  2.09973946e-03,  1.65916158e-02,  5.48514592e-02,
        1.26292341e-01,  2.37553208e-01,  3.91878885e-01,  5.88735053e-01,
        8.23686220e-01,  1.08854725e+00,  1.37180080e+00,  1.65925535e+00,
        1.93490221e+00,  2.18191685e+00,  2.38373956e+00,  2.52516524e+00,
        2.59337087e+00,  2.57881293e+00,  2.47593511e+00,  2.28363857e+00,
        2.00548243e+00,  1.64959896e+00,  1.22832697e+00,  7.57584262e-01,
        2.56017602e-01, -2.56017602e-01, -7.57584262e-01, -1.22832697e+00,
       -1.64959896e+00, -2.00548243e+00, -2.28363857e+00, -2.47593511e+00,
       -2.57881293e+00, -2.59337087e+00, -2.52516524e+00, -2.38373956e+00,
       -2.18191685e+00, -1.93490221e+00, -1.65925535e+00, -1.37180080e+00,
       -1.08854725e+00, -8.23686220e-01, -5.88735053e-01, -3.91878885e-01,
       -2.37553208e-01, -1.26292341e-01, -5.48514592e-02, -1.65916158e-02,
       -2.09973946e-03,  0.00000000e+00])
```

In [9]:
```
fig0 = plt.figure()
ax0 = fig0.add_subplot()

ax0.plot(x, y)
```

Out[9]:
```
[<matplotlib.lines.Line2D at 0x7dd449737a70>]
```

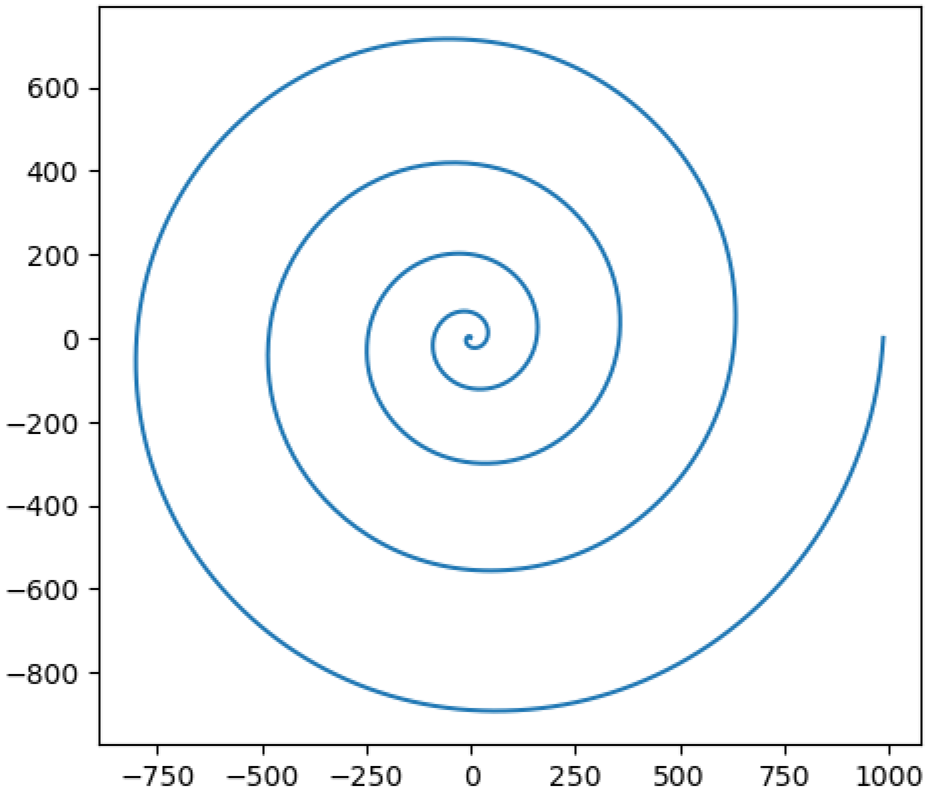


part b

In [40]:
```
theta = np.linspace(0,10*np.pi, 1000)

r = theta**2

x = r*np.cos(theta)
y = r*np.sin(theta)
```

In [41]:
```
fig1 = plt.figure()
ax1 = fig1.add_subplot(aspect="equal")


ax1.plot(x,y)
```

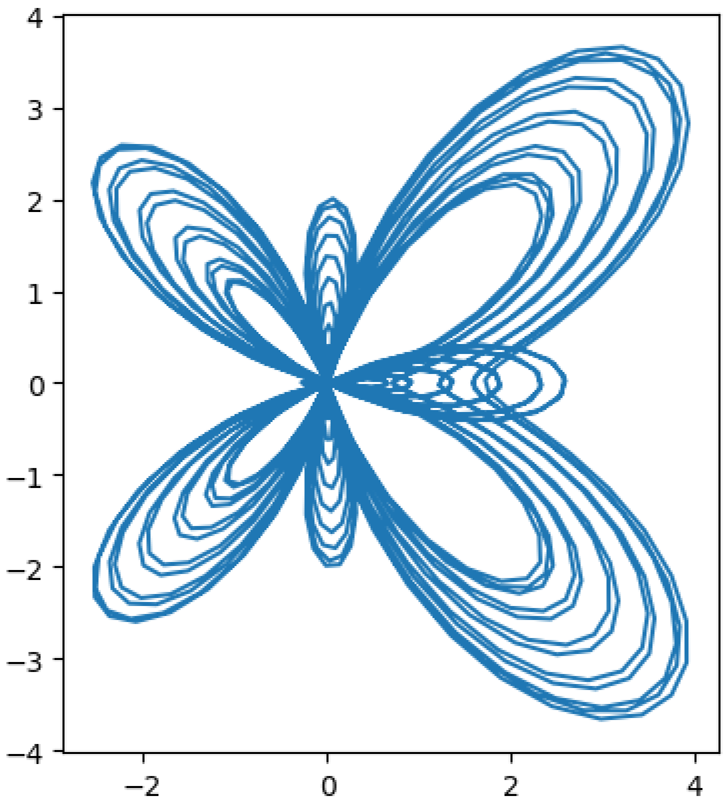
Out[41]:
```
[<matplotlib.lines.Line2D at 0x7dd43fb384d0>]
```



part c

In [42]:
```
theta = np.linspace(0,24*np.pi, 1000)

r = np.exp(np.cos(theta))-2*np.cos(4*theta)+np.sin(theta/12)

x = r*np.cos(theta)
y = r*np.sin(theta)
```

In [43]:
```
fig2, ax2 = plt.subplots(subplot_kw = {'aspect':'equal'})

ax2.plot(x, y)
plt.show()
```



In [ ]: